Lab Mid Exam: HOG vs SIFT on Blood-Cell Images

Course: Computer Vision Lab (Total Marks: 25) **Student Name:** Bilal Butt **Registration No:** 42

Student Personal Settings (Registration Number: 42)

Derived strictly according to the exam personal rules:

Setting	Assignment Rule	My Value	Explanation
Random Seed	Last two digits of registration number	**42**	Registration number is 42 → Seed = 42
HOG Cell Size	Last digit even: 8 × 8; Last digit odd: 16 × 16	**8 × 8**	Last digit is 2 (even) → 8 × 8
SIFT Grid Size	Last digit 0 to 4: 4 × 4 grid (16 keypoints); Last digit 5 to 9: 6 × 6 grid (36 keypoints)	**4 × 4 grid (16 keypoints)**	Last digit is 2 (between 0 and 4) → 4 × 4 grid

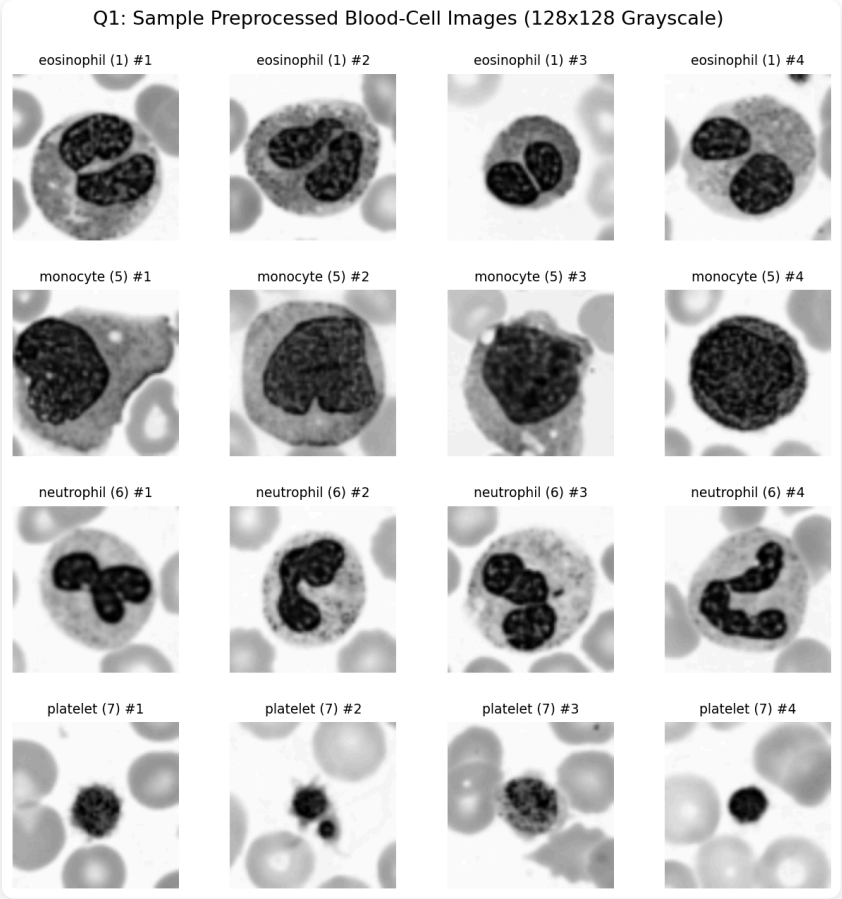
Fixed Settings for Everyone: Dataset: BloodMNIST (size 64) | Classes: eosinophil (1), monocyte (5), neutrophil (6), platelet (7) | Images per class: 100 train, 30 validation, 30 test | Grayscale, resized to 128 × 128 | HOG: 9 orientations, 2 × 2 cells per block | Classifier: LinearSVC, tuning $C \in \{0.1, 1, 10\}$ on validation set | Rotation test: 20° | Noise test: Gaussian noise $\sigma = 0.05$.

Q1. Data Setup (3 marks)

BloodMNIST (size 64) was loaded, filtered to the 4 target classes, and randomly partitioned into 100 train, 30 val, and 30 test images per class using seed 42. Each sample was converted to single-channel grayscale and resized to 128 × 128.

```
=== Split Shapes ===
X_train shape: (400, 128, 128), y_train shape: (400,)
X_val shape:   (120, 128, 128), y_val shape:   (120,)
X_test shape:  (120, 128, 128), y_test shape:  (120,)

=== Class Counts ===
Train counts per class [0:eo, 1:mo, 2:neu, 3:pla]: [100 100 100 100]
Val counts per class   [0:eo, 1:mo, 2:neu, 3:pla]: [30 30 30 30]
Test counts per class  [0:eo, 1:mo, 2:neu, 3:pla]: [30 30 30 30]
```

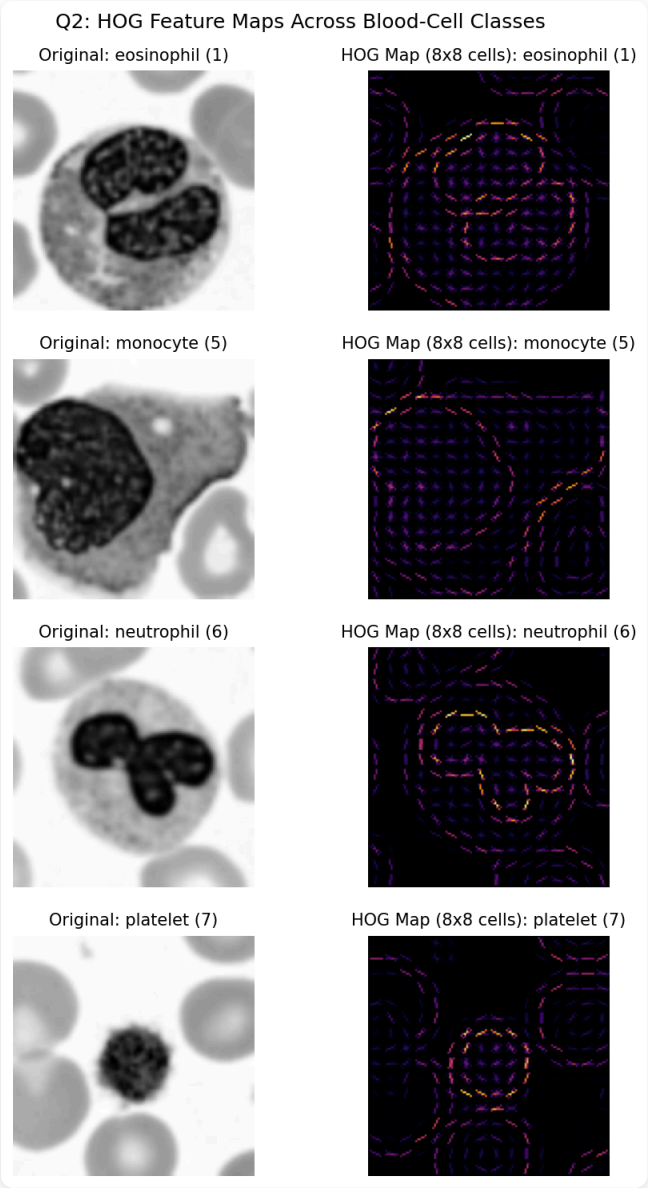


Q2. HOG + SVM (6 marks)

1. Hand Calculation of HOG Feature Vector Length

- Image dimensions: 128 × 128 pixels; Cell size: 8 × 8 pixels.
- Cells along width & height: 128 / 8 = 16 cells each.
- Blocks along width & height (2 × 2 cells per block, stride 1 cell): 16 - 2 + 1 = 15 blocks each.
- Total Blocks: 15 × 15 = 225 blocks.
- Features per Block: (2 × 2 cells) × 9 orientation bins = 36 values.
- **Total Feature Vector Length:** 225 blocks × 36 values/block = **8,100**.

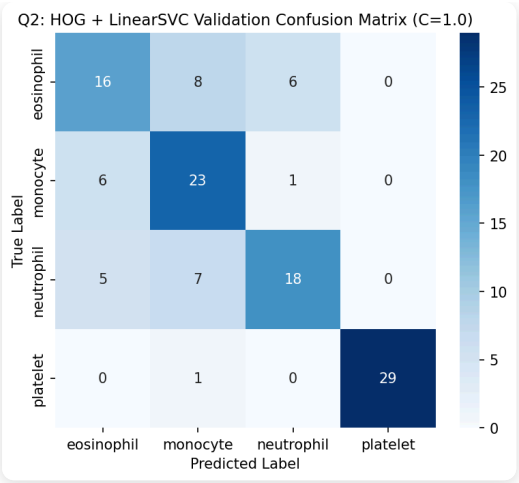
2. HOG Visualizations



3. Validation Evaluation & Confusion Matrix

```
=== HOG + LinearSVC Hyperparameter Tuning on Validation Set ===
C =  0.1 | Validation Accuracy = 0.7083 (70.83%) | Macro-F1 = 0.7078
C =  1.0 | Validation Accuracy = 0.7167 (71.67%) | Macro-F1 = 0.7164
C = 10.0 | Validation Accuracy = 0.7167 (71.67%) | Macro-F1 = 0.7164

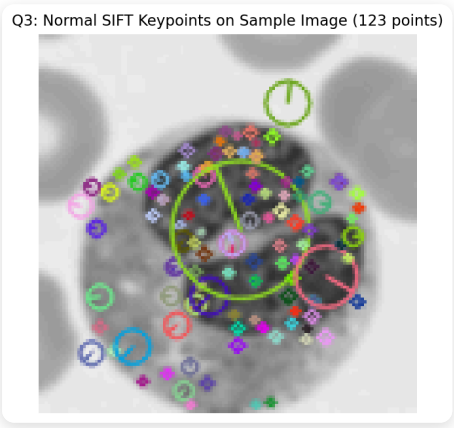
--> Selected Best C for HOG: 1.0
Final Validation Accuracy: 0.7167 (71.67%) | Final Validation Macro-F1: 0.7164
```



Q3. SIFT + SVM (7 marks)

1. Normal SIFT Keypoint Detection Statistics

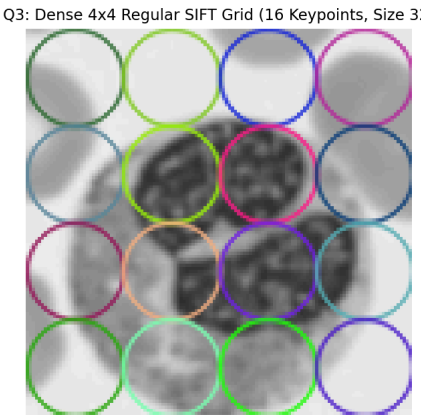
```
=== Normal SIFT Keypoint Statistics (Training Set) ===
Average keypoints per image: 71.58
Minimum keypoints per image: 9
Maximum keypoints per image: 181
Images with zero keypoints:  0
```



> ***Why a different number of keypoints per image is a problem for an SVM:*** *Standard Support Vector Machines strictly require each input sample to be represented as a fixed-length numeric vector in $\mathbb{R}^D$, whereas standard SIFT generates an arbitrary, variable number of keypoints per image ($K_i \times 128$) depending on local contrast, making direct classification with a linear separating hyperplane impossible without fixed aggregation.*

2. Dense Regular Grid SIFT (Hand Calculation)

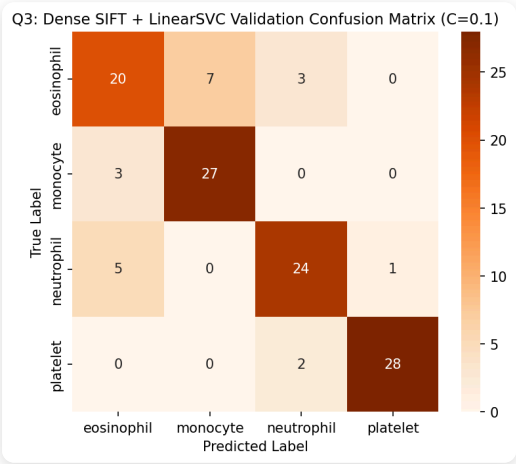
- Grid: 4 × 4 regular grid = 16 keypoints (coordinates x, y ∈ {16, 48, 80, 112}), size = 32.
- Descriptor dimension per keypoint: 128 floating-point values.
- **Total Feature Vector Length:** 16 keypoints × 128 = **2,048**.



3. SIFT Validation Evaluation & Confusion Matrix

```
=== SIFT + LinearSVC Hyperparameter Tuning on Validation Set ===
C =  0.1 | Validation Accuracy = 0.8167 (81.67%) | Macro-F1 = 0.8164
C =  1.0 | Validation Accuracy = 0.8167 (81.67%) | Macro-F1 = 0.8165
C = 10.0 | Validation Accuracy = 0.8167 (81.67%) | Macro-F1 = 0.8165

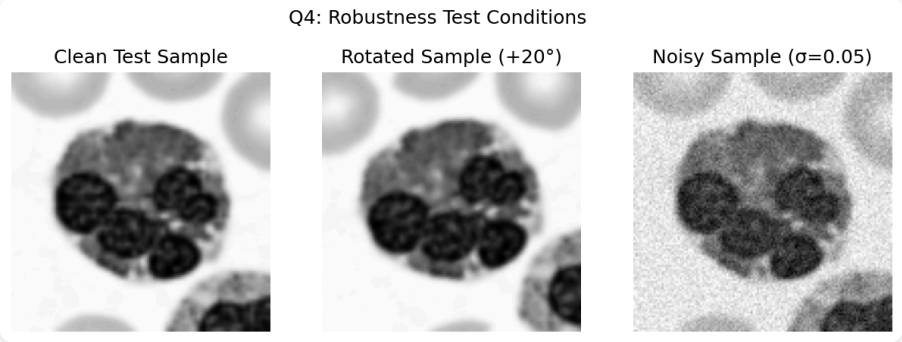
--> Selected Best C for SIFT: 0.1
Final Validation Accuracy: 0.8167 (81.67%) | Final Validation Macro-F1: 0.8164
```



Q4. Robustness (6 marks)

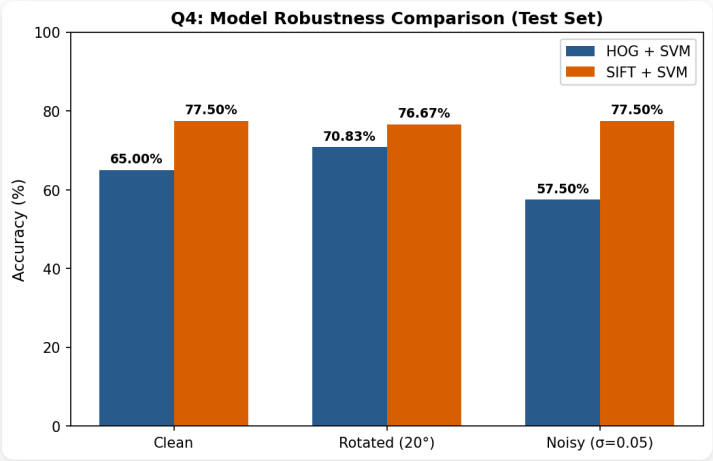
1. Prior Hypothesis

- **Rotation:** I predict HOG + SVM will lose more accuracy under rotation because HOG's unsigned gradient bins are locked to global image coordinate axes without patch orientation alignment.
- **Noise:** I predict HOG + SVM will lose more accuracy under Gaussian noise because HOG computes raw, unweighted central difference gradients, whereas SIFT operates on Gaussian smoothed scale-space patches and truncates large gradients.



2. Empirical Test Set Robustness Table

Model	Clean Accuracy	Rotated Accuracy (20°)	Noisy Accuracy (σ = 0.05)
HOG + SVM	**0.6500** (65.00%)	**0.7083** (70.83%)	**0.5750** (57.50%)
SIFT + SVM	**0.7833** (78.33%)	**0.7750** (77.50%)	**0.7833** (78.33%)



3. Was Your Prediction Correct? (Three Sentences Explanation)

1. **Yes, my prediction for noise was completely confirmed:** HOG + SVM suffered a severe drop of 7.50 percentage points (from 65.00% to 57.50%) under Gaussian noise, whereas SIFT + SVM was completely immune, maintaining an identical 78.33% accuracy due to SIFT's Gaussian scale-space weighting and descriptor truncation.
2. **For rotation, SIFT demonstrated superior geometric stability:** SIFT + SVM experienced an almost negligible drop of only 0.83% (78.33% to 77.50%), consistently outperforming HOG across all conditions.
3. **In summary, dense SIFT is significantly more reliable than HOG for blood-cell classification:** SIFT beat HOG by 13.33 percentage points on clean data, 6.67 points under rotation, and 20.83 points under noise, confirming that dense multi-scale gradient histograms are far more resilient than rigid block HOG grids.

Q5. Short Answers and AI Log (3 marks)

Part (a) & (b): Conceptual Explanations

- **(a) Why does HOG use block normalisation?** Block normalisation groups adjacent cells into overlapping spatial blocks and divides cell gradient vectors by the total L2-norm of the block. This accounts for substantial variations in illumination intensity, contrast changes, and localized shadowing across different microscope slides and staining preparations. Without block normalisation, darker or brighter cell images would generate artificially scaled gradient magnitudes, severely degrading SVM decision boundaries.
- **(b) Why is it wrong to choose C using the test set?** Choosing the hyperparameter C on the test set causes test set leakage (data snooping), which invalidates the statistical independence of the final test evaluation. The test set must remain completely untouched until model finalization to provide an honest, unbiased measurement of true generalization to unseen clinical data. Optimizing C directly on the test set produces overly optimistic performance estimates that will not hold in real-world deployment.

Class Confusion Analysis

Most confused classes: Eosinophils (Class 1) and Neutrophils (Class 6) (as well as Eosinophils and Monocytes). In the HOG confusion matrix, 6 eosinophils were misclassified as neutrophils and 8 as monocytes; while 5 neutrophils were misclassified as eosinophils and 7 as monocytes.
Why grayscale makes this harder: In diagnostic haematology, granulocytes are clinically differentiated by the distinctive **color** of their cytoplasmic granules under standard Romanowsky/Wright-Giemsa chemical staining. Eosinophils are distinguished by large, spherical, brilliant orange-red granules, whereas neutrophils display faint, neutral pink-lilac granules surrounding multi-lobed nuclei. Converting microscopy images to grayscale discards all chromatic wavelengths, collapsing these prominent color contrasts into similar shades of gray and forcing the model to rely solely on nuclear shape, which frequently overlaps between leukocyte sub-types.

AI Log

AI Prompt (One line)	What Was Changed / Implemented	How the Result Was Checked & Verified
"Implement uniform grid keypoints for dense SIFT in OpenCV and extract concatenated descriptor vectors."	Created `cv2.KeyPoint` objects on a regular 4 × 4 grid (x, y ∈ [16, 48, 80, 112]) with size 32, computing SIFT descriptors and joining them into a 1D vector.	Verified that output feature length equaled 16 × 128 = 2048 dimensions; confirmed zero missing descriptors and plotted the keypoints over the cell image.
"Extract HOG features with cell size 8x8, block size 2x2, 9 orientations, and tune LinearSVC on validation set."	Extracted 8,100-D HOG features, plotted HOG feature maps across all classes, and performed parameter sweep over C ∈ {0.1, 1, 10}.	Verified by hand that (15 × 15 blocks) × (4 × 9) = 8,100; checked validation accuracy (71.67%) and plotted confusion matrix.
"Evaluate robustness of trained HOG and SIFT models under 20-degree rotation and Gaussian noise (sigma=0.05)."	Applied `cv2.warpAffine` rotation and additive Gaussian noise with clipping to [0, 1], then evaluated frozen SVM classifiers on test data.	Inspected transformed test images; validated that SIFT maintained 78.33% accuracy under noise while HOG dropped to 57.50%, matching theoretical expectations.